Year 13 Further Maths - Work, energy and power

Similar questions - Vector methods

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The coordinates of A and B are $(-2,1)$ and $(6,7)$ (units are metres). A particle P moves from A to B under several forces, one of which is $\mathbf{F} = (3\mathbf{i} - 2\mathbf{j})$ N.

(a) Find the work done by $\mathbf{F}$. **[2]**

(b) At B the velocity of P is $(-\mathbf{i} + 3\mathbf{j})\ \mathrm{m\,s^{-1}}$. Find the power generated by $\mathbf{F}$ and say what the sign means. **[3]**

(c) A light elastic string of natural length 7 m and modulus 28 N joins the origin to P. Find the tension and elastic energy when P is at B. **[4]**

2 A particle Q of mass m kg moves with constant acceleration $(2\mathbf{i} - \mathbf{j})\ \mathrm{m\,s^{-2}}$. Initially Q is at O with velocity $(-3\mathbf{i} + 4\mathbf{j})\ \mathrm{m\,s^{-1}}$. After 4 seconds it is at A.

(a) Use $\mathbf{v} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{u} + 2\mathbf{a} \cdot \mathbf{x}$ to show that the kinetic energy at A is $12.5m$ J. **[5]**

(b) Show that the power generated initially is $-10m$ W and say what this means. **[3]**

(c) Find the time at which the power generated is instantaneously zero, and the minimum kinetic energy. **[4]**

3 A particle of mass 2 kg moves on a smooth horizontal surface with velocity $\mathbf{v} = 12\sinh t\,\mathbf{i} + (12\cosh t - 13)\mathbf{j}\ \mathrm{m\,s^{-1}}$ at time t seconds, $t \geqslant 0$, acted upon by a variable horizontal force $\mathbf{F}$.

(a) By considering kinetic energy, find the work done by $\mathbf{F}$ over $0 \leqslant t \leqslant \ln 3$. **[5]**

(b) Find the rate at which $\mathbf{F}$ is working when the particle is moving parallel to $\mathbf{i}$. **[6]**

Question 1

(a) $\mathbf{d} = (8,6)$, $W = 3(8) - 2(6) = 12$ J

(b) $P = \mathbf{F} \cdot \mathbf{v} = -3 - 6 = -9$ W; $\mathbf{F}$ is opposing the motion (removing energy)

(c) Length $OB = \sqrt{85} = 9.22$, extension 2.22 m. $T = \dfrac{28(2.22)}{7} = 8.88$ N; EPE $= \dfrac{28(2.22)^2}{14} = 9.85$ J

Question 2

(a) $\mathbf{x} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 = (4,8)$; $\mathbf{u} \cdot \mathbf{u} = 25$, $\mathbf{a} \cdot \mathbf{x} = 0$ so $\mathbf{v} \cdot \mathbf{v} = 25$ and KE $= \frac{1}{2}m(25) = 12.5m$

(b) $P = m\mathbf{a} \cdot \mathbf{u} = m(2(-3) - 1(4)) = -10m$; the force initially removes energy from Q (it is slowing down)

(c) $\mathbf{v} = (-3+2t, 4-t)$; $\mathbf{a} \cdot \mathbf{v} = 2(-3+2t) - (4-t) = 5t - 10 = 0$, $t = 2$. Then $\mathbf{v} = (1,2)$, minimum KE $= \frac{1}{2}m(5) = 2.5m$ J

Question 3

(a) At $t = 0$, $\mathbf{v} = (0,-1)$, KE $= 1$ J. At $t = \ln 3$, $\sinh = \frac{4}{3}$, $\cosh = \frac{5}{3}$, $\mathbf{v} = (16,7)$, KE $= \frac{1}{2}(2)(305) = 305$ J. Work done $= 305 - 1 = 304$ J

(b) Parallel to $\mathbf{i}$: $12\cosh t = 13$, $\cosh t = \frac{13}{12}$, $\sinh t = \frac{5}{12}$; $\mathbf{v} = 5\mathbf{i}$. $\mathbf{a} = (12\cosh t, 12\sinh t) = (13,5)$, $\mathbf{F} = (26,10)$; power $= \mathbf{F} \cdot \mathbf{v} = 130$ W